Matthew Kim

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Education

University of Virginia Charlottesville, VA
B.A. in Applied Statistics, Minor in Computer Science | GPA: 3.9/4.0 *Aug. 2023 – May 2027*

- **Relevant Coursework:** Statistical Machine Learning, Data Visual and Management, Data Analysis with Python, Design & Analysis of Sample Surveys, Intro Mathematical Statistics, From Data to Knowledge, Intro to Regression Analysis, Software Dev Essentials, Computer Systems and Org 1, Discrete Math and Theory 1, Data Structures and Algo 1

Experience

Research Assistant 2024 – Present
UVA Historic Fashion Archive *Charlottesville, VA*

- Architected a Next.js 16 / React 19 / TypeScript frontend for UVA's historic clothing collection on a CollectiveAccess CMS (custom API client, 5-minute cache, static JSON fallback), with multi-attribute filtering, Boolean search, JSON/CSV/PDF export, and a Three.js 3D garment viewer
- Rebuilt the related-item recommender as a learned re-ranker over FashionCLIP image embeddings and extracted attributes, raising nDCG@5 from 0.70 to 0.93 (95% CI +0.15 to +0.31) on a 50-query blind-judged benchmark under nested cross-validation
- Deployed collection search fusing BM25 with FashionCLIP text-to-image retrieval as a CPU-only FastAPI service (nDCG@5 0.80 vs. 0.68 for keyword search), and prototyped a local Qwen3-8B query parser with citation-checked RAG answers (nDCG@5 0.88); gated by GitHub Actions and tracked in MLflow
- Extracted material, color, and technique attributes from 1,400+ catalog records (F1 0.95–0.99) and trained a linear probe on weak labels to tag undescribed garments from photos (color F1 0.58 vs. 0.39 zero-shot)
- Piloted a 3D shape-descriptor pipeline (curvature, symmetry, drape via `trimesh`) on photogrammetry scans, applying PCA and clustering evaluated against catalog metadata (ARI, NMI)

Teaching Assistant 2026 – Present
University of Virginia — CS1112 *Charlottesville, VA*

- Supported 300+ students in introductory Python through weekly office hours, lab sections, and one-on-one debugging
- Graded assignments and exams with structured feedback on data structures, control flow, and functions

Projects

Hana-Bi: E-Commerce Platform | *Next.js, TypeScript, Stripe, Resend, Zustand, Framer Motion* 2024 – Present

- Engineered a full-stack e-commerce platform with Stripe as the product data source of truth, implementing an autopaging catalog pipeline, a per-product metadata schema, and serverless-safe session verification via direct Stripe API calls
- Built an end-to-end transactional data flow: client cart state (Zustand, persisted locally), Checkout Session creation, webhook-driven order confirmation emails via Resend, and a waitlist capture pipeline with dual-notification routing
- Structured a multi-environment deployment checklist covering API key rotation, webhook secrets, and sender domain verification

Tech Industry Layoff Analysis | *R, tidyverse, ggplot2, R Markdown* 2026

- Analyzed 2,700+ tech layoffs using R and tidyverse, applying statistical modeling techniques including t-tests and ANOVA to compare workforce reduction severity across company types and industries
- Built exploratory visualizations in ggplot2 and engineered features from raw data; evaluated workforce reduction patterns and found that private companies cut proportionally twice as much of their workforce as public ones (32.4% vs. 16.7%)

Insurance Client Data Management System | *Python, Flask, SQLite, Google Sheets API, Render* 2025 – 2026

- Built a Flask dashboard that syncs the agency's Google Sheet into a read-only SQLite copy for search, sorting, and CSV export, replacing a binder-based workflow covering ∼2,000 records; rebuilt in 2026 with a 30/60/90-day renewals view, an audit log, and a pytest suite in GitHub Actions

NBA Salary Prediction Model | *R, multiple linear regression, cross-validation* 2025

- Built a multiple linear regression model of NBA player salaries (adj. $R^2 \approx 0.62$, matched by five-fold cross-validation), with full EDA, variable selection, assumption diagnostics, and Cook's distance leverage checks; Points, Minutes, Stocks, and Age were the key predictors

Technical Skills

Languages: Python, R, Java, C, JavaScript, TypeScript, SQL
Machine Learning & AI: PyTorch, scikit-learn, OpenCLIP (FashionCLIP), Ollama, MLflow, RAG, Generative AI, retrieval & ranking (BM25, nDCG), MLOps/LLMOps
Frameworks & Tools: React, Next.js, FastAPI, Flask, Tailwind CSS, pandas, NumPy, SciPy, tidyverse, ggplot2, plotly, R Shiny, rvest (web scraping), Three.js, Git/GitHub, GitHub Actions (CI/CD), Stripe API, REST APIs, Jupyter
Statistical Methods: Hypothesis testing, ANOVA, linear & logistic regression, ridge/lasso regularization, KNN, LDA, decision trees, splines, bootstrap, Monte Carlo simulation, cross-validation, ROC/AUC, precision/recall, EDA, Optimization, Forecasting, Model Evaluation
Survey Sampling: Stratified & cluster sampling, sample-size determination, ratio & regression estimation, capture-recapture